1.1 Right-angle triangle with numbers:

```
// Name: Utkarsha Patil
// Roll no: SEA52
// Right-angle triangle with numbers

#include<stdio.h>
void main(){
    int n = 4;
    for(int i = 1; i <= n; i++){
        for(int j = 1; j <= i; j++){
            printf ("%d ", j);
        }
        printf("\n");
    }
}
```

Output:

```
1
1 2
1 2 3
1 2 3 4
```

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1.2 Diamond shape with numbers:

```
// Name: Utkarsha Patil
// Roll no: SEA52
// Diamond shape with numbers

#include<stdio.h>
void main(){
    int n = 4;
    for(int i = 1; i <= n; i++){
        for(int j = n-1; j >= i; j--){
            printf(" ");
        }
        for(int j = i; j > 0; j--){
            printf("%d ", i);
        }
        printf("\n");
    }
    for(int i = n-1; i > 0; i--){
        for(int j = i; j < n; j++){
            printf(" ");
        }
        for(int j = 0; j < i; j++){
            printf("%d ", i);
        }
        printf("\n");
    }
}
```

Output:

```
   1
  2 2
 3 3 3
4 4 4 4
 3 3 3
  2 2
   1
```

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1.3 Pyramid with asterisks:

```
// Name: Utkarsha Patil
// Roll no: SEA52
// Pyramid with asterisks

#include<stdio.h>
void main(){
   int n = 4;
   for(int i = 1; i <= n; i++){
      for(int j = n-1; j >= i; j--){
         printf(" ");
      }
      for(int j = i; j > 0; j--){
         printf("* ");
      }
      printf("\n");
   }
}
```

Output:

```
   *
  * *
 * * *
* * * *
```

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1.4 Pyramid using alphabets:

```
// Name: Utkarsha Patil
// Roll no: SEA52
// Pyramid using alphabets

#include<stdio.h>
void main(){
    int n = 4;
    for(int i = 0; i < n; i++){
        char ch = 'A';
        for(int j = n-1; j > i; j--){
            printf(" ");
        }
        for(int j = 0; j <= i; j++){
            printf("%c", ch+j);
        }
        for(int j = i; j > 0; j--){
            printf("%c", ch+j-1);
        }
        printf("\n");
    }
}
```

Output:

```
   A
  ABA
 ABCBA
ABCDCBA
```

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2. Student Result Database Management using Array of Structures

```
// Name: Utkarsha Patil
// Roll no: SEA52
// Student Result Database Management using Array of Structures

#include<stdio.h>

typedef struct stud{
   int rn;
   char name[20];
   char program[20];
   char sub1[10], sub2[10];
   int sub1_mk, sub2_mk, total;
   float avg;
}stud;

void main(){
   stud s[20];
   int n, op, key;
   void create(stud[], int);
   void display(stud[], int);
   void search(stud[], int, int);
   printf("STUDENTS DATABASE MANAGEMENT\n");

   do{
      printf("\n1. Create\n2. Display\n3. Search\n");
      printf("Enter option no. : ");
      scanf("%d", &op);
      switch(op){
         case 1:
      printf("\nEnter no. of students: ");
      scanf("%d", &n);
      create(s, n);
      break;
      case 2:
      display(s, n);
      break;
      case 3:
      search(s, n, key);
      break;
   }
}while(op < 3);
```

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```
}

void create(stud s[], int n){
   printf("\n---//ENTER STUDENTS DATA//---\n");
   for(int i = 1; i <= n; i++){
      printf("\n~~~Student %d Data~~~\n", i);
      printf("Enter student's name: ");
      scanf("%s", &s[i].name);
      printf("Enter student's roll no. : ");
      scanf("%d", &s[i].rn);
      printf("Enter student's program: ");
      scanf("%s", &s[i].program);
      printf("Enter subject1: ");
      scanf("%s", &s[i].sub1);
      printf("Enter subject2: ");
      scanf("%s", &s[i].sub2);
      printf("Enter subject 1's marks: ");
      scanf("%d", &s[i].sub1_mk);
      printf("Enter subject 2's marks: ");
      scanf("%d", &s[i].sub2_mk);

      s[i].total = s[i].sub1_mk + s[i].sub2_mk;
      s[i].avg = s[i].total/2;
   }
}

void display(stud s[], int n){
   printf("\n---//DISPLAYING STUDENTS DATA//---");
   for(int i = 1; i <= n; i++){
      printf("\n\n~~~Student %d Data~~~", i);
      printf("\nName: %s", s[i].name);
      printf("\nRoll no. : %d", s[i].rn);
      printf("\nProgram: %s", s[i].program);
      printf("\nSubject 1: %s", s[i].sub1);
      printf("\nSubject 2: %s", s[i].sub2);
      printf("\nSubject 1's marks: %d", s[i].sub1_mk);
      printf("\nSubject 2's marks: %d", s[i].sub2_mk);
      printf("\nTotal: %d", s[i].total);
      printf("\nAverage: %.2lf", s[i].avg);
   }
   printf("\n\n---//END//---\n");
}

void search(stud s[], int n, int key){
```

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```
    int flag = 0;
    printf("\nEnter student's roll no. to search: ");
    scanf("%d", &key);
    for(int i = 1; i <= n; i++){
        if(s[i].rn == key){
            printf("STUDENT FOUND\n");
            flag++;
            break;
        }
    }
    if(flag == 0){
        printf("STUDENT NOT FOUND\n");
    }
}
```

Output:

```
STUDENTS DATABASE MANAGEMENT

1. Create
2. Display
3. Search
Enter option no. : 1

Enter no. of students: 2

---//ENTER STUDENTS DATA//---

~~~Student 1 Data~~~
Enter student's name: Pushpak
Enter student's roll no. : 13
Enter student's program: EnTC
Enter subject1: DSA
Enter subject2: DE
Enter subject 1's marks: 90
Enter subject 2's marks: 95

~~~Student 2 Data~~~
Enter student's name: Neel
Enter student's roll no. : 9
Enter student's program: EnTC
Enter subject1: DSA
Enter subject2: DE
```

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```
Enter subject 1's marks: 98
Enter subject 2's marks: 100

1. Create
2. Display
3. Search
Enter option no. : 2

---//DISPLAYING STUDENTS DATA//---

~~~Student 1 Data~~~
Name: Pushpak
Roll no. : 13
Program: EnTC
Subject 1: EC
Subject 2: DE
Subject 1's marks: 90
Subject 2's marks: 95
Total: 185
Average: 92.00

~~~Student 2 Data~~~
Name: Neel
Roll no. : 9
Program: EnTC
Subject 1: EC
Subject 2: DE
Subject 1's marks: 98
Subject 2's marks: 100
Total: 198
Average: 99.00

---//END//---

1. Create
2. Display
3. Search
Enter option no. : 3

Enter student's roll no. to search: 15
STUDENT NOT FOUND
```

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4. Decimal to Binary Conversion using Stack

```
// Name: Utkarsha Patil
// Roll no: SEA52
// Decimal to Binary Conversion using Stack

#include<stdio.h>
void main(){
   int n;
   printf("Enter Decimal number: ");
   scanf("%d", &n);
   int s[10], top = -1, r;

   while(n != 0){
      r = n%2;
      n = n/2;
      top++;
      s[top] = r;
   }

   printf("Binary number for given decimal number is: ");
   for(int i = top; i >= 0; i--){
      printf("%d", s[i]);
   }
}
```

Output:

```
Enter Decimal number: 15
Binary number for given decimal number is: 1111
```

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